(Please write your Exam Roll No.) Exam Roll No. .....

# END TERM EXAMINATION

THIRD SEMESTER [B.TECH] JANUARY 2024

**Paper Code: ECC-205** **Subject: Signals and Systems**

**Time: 3 Hours** **Maximum Marks: 75**

***Note: Attempt five questions in all including Q. No.1 which is compulsory. Select one question from each unit. Assume missing data if any.***

Q1 Attempt all questions:-

(a) Classify the Following signals a1) Even and Odd a2) Periodic and Aperiodic a3) Energy and Power signals **(3)**

(b) Define the properties of ROC for Laplace Transform. **(3)**

(c) Calculate number of additions and multiplications for N point DFT and FFT for N=16. **(3)**

(d) Define the Causality and Stability criteria using Z Transform. **(3)**

(e) Define Parseval's Relation. **(3)**

**UNIT-I**

Q2 (a) Describe impulse, step and ramp signals. **(3)**

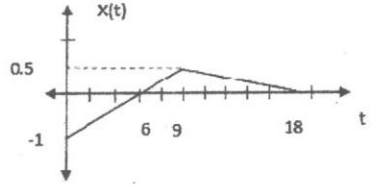
(b) Check whether energy or power signal: **(3)**

B1) $x(t) = e^{j(2t+\frac{\pi}{4})}$ B2) $x(n) = \cos\{(\Pi/4)n\}$

(c) Find the fundamental period of following signals. **(3)**

C1) x(t) = cos(10t+1) – sin(4t-1) C2) x(n) = sin(3Πn + 0.6Πn)

(d) Perform the following transformations on given signal. **(6)**




a) x(-t) b) x(t +6) c) x(3t) d) x(3t + 6)

Q3 (a) Find the Fourier Series Coefficients of the Following Function: **(5)**

$x(t) = 1 + \sin(\omega_0 t) + 2\cos(\omega_0 t) + \cos(2\omega_0 t + \Pi/4)$

(b) Find the Fourier series Coefficients for: **(10)**

1) $x(n) = \cos\sqrt{2}\,\pi n$ 2) $x(n) = \cos \pi n/3$ 3) x(n) = [ 1 1 0 0]

**UNIT-II**

Q4 (a) Find the Fourier Transform of a signal $x(t) = \frac{2}{1+t^2}$. However, the Fourier Transform of $g(t) = e^{-a|t|}$ is given as $G(\omega) = \frac{2a}{1+\omega^2}$ **(7.5)**

(b) An LTI system is described by the differential equation **(7.5)**

$\frac{dy(t)}{dx} + (t) = (t)$, Find the impulse response of the system

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Q5 (a) Determine whether system y(n) = x(n+5) – x(n-7) is linear or not. **(5)**

(b) Determine whether system $y(t) = t^2x(t-4)$ is time invariant. **(5)**

(c) Find the Laplace Transform of the following Signals: **(5)**

1) Unit Step Signal u(t) 2) Delayed Unit Step Ku(t-a)

3) Ramp Signal r(t)

**UNIT-III**

Q6 An 8 – point sequence is given by x(n) = [ 2, 2, 2, 2, 1, 1, 1, 1]. Compute 8 point DFT of x(n) by radix 2 DIF FFT. **(15)**

Q7 (a) Perform the circular convolution of following sequences: **(7.5)**

x(n) = [1 2 2 3] and h(n) = [ 1 3 2 1]

(b) Perform the N point DFT on the following sequence: **(7.5)**

x(n) = [ 1, 2, 3, 3] and also plot magnitude & Phase plots.

**UNIT-IV**

Q8 (a) Define any Four properties of Z Transforms. **(7.5)**

(b) Find the Z transform of Following Signals: **(7.5)**

1) $x(n) = 2^nU(n)$ 2) $x(n) = -2^n U[-n-1]$

Q9 Find the Inverse Z Transform of the function $X(Z) = \frac{1}{1-1.5Z^{-1}+0.5Z^{-2}}$

by using 1) Long Division method 2) Partial Fraction Expansion, **(15)**

a) when ROC: |Z| > 1 b) when ROC |Z| < 1/2

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Total Number of Pages: 01

Student Name :_____

Enrollment Number:_____

**Mid Term Examination- Nov 2022**

Program : B. Tech
Paper Code : ECC 205
Time Duration : $1\frac{1}{2}$ hours

Semester : Third
Paper Name : Signals and Systems
Maximum Marks : 30

**Note:** Question No. 1 is compulsory. Attempt any two questions from the remaining questions. Only scientific calculator is allowed.

## Question 1

**1(a)** Check whether the system described by $y(n) = x(n) + 6$ is linear or non-linear. [2]

**1(b)** Sketch $x(-2t+1)$ for a given signal $x(t) = 2[u(t) - u(t-3)]$. [2]

**1(c)** Determine whether a signal $x(t) = e^{-a|t|}$, $a > 0$ is energy or power signal or neither. [2]

**1(d)** Determine the Nyquist rate of a signal $x(t) = \frac{sin(4000\pi t)}{\pi t}$. [2]

**1(e)** Obtain the Fourier series coefficients of a discrete time signal $x(n) = 0.5 + cos(3\pi n + \pi/3)$. [2]

## Question 2

**2(a)** Find the output of an LTI systems whose impulse response is $h(t) = e^{-2t}u(t)$ and input is $x(t) = e^{-t+1}u(t-1)$. [5]

**2(b)** Find the linear convolution of two sequences $x_1(n) = [2, \underset{\uparrow}{1}, -1, 3]$ and $x_2(n) = [\underset{\uparrow}{2}, 1, 5]$. [5]

## Question 3

**3(a)** Determine a signal $x(t)$, when the following facts about $x(t)$ are given:

1. $x(t)$ is a real and odd signal.
2. $x(t)$ is periodic with period $T = 4$, and it has exponential Fourier series coefficients $c_k$.
3. $c_k = 0$ for $|k| > 1$.
4. $\frac{1}{4}\int_0^4 |x(t)|^2 dt = 4$.

[5]

**3(b)** Consider a causal LTI system with frequency response $H(\omega) = \frac{1}{3+j\omega}$. For a particular input $x(t)$, this system is observed to produce the output $y(t) = e^{-3t}u(t) - e^{-4t}u(t)$. Determine the input $x(t)$. [5]

## Question 4

**4(a)** Find the Fourier transform of a signal $x(t) = \frac{1}{1+t^2}$. However, the Fourier transform of $g(t) = e^{-|t|}$ is given as: $G(\omega) = \frac{2}{1+\omega^2}$. [5]

**4(b)** The Laplace transform of a signal $x(t)$ is given as $X(s) = \frac{3s+1}{s^3+4s^2+(k-3)s}$. If $\lim_{t\to\infty} x(t) = 1$, then find the value of $k$ ? [5]